



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SHRI SHIRIDI SAI SERVICE STN

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

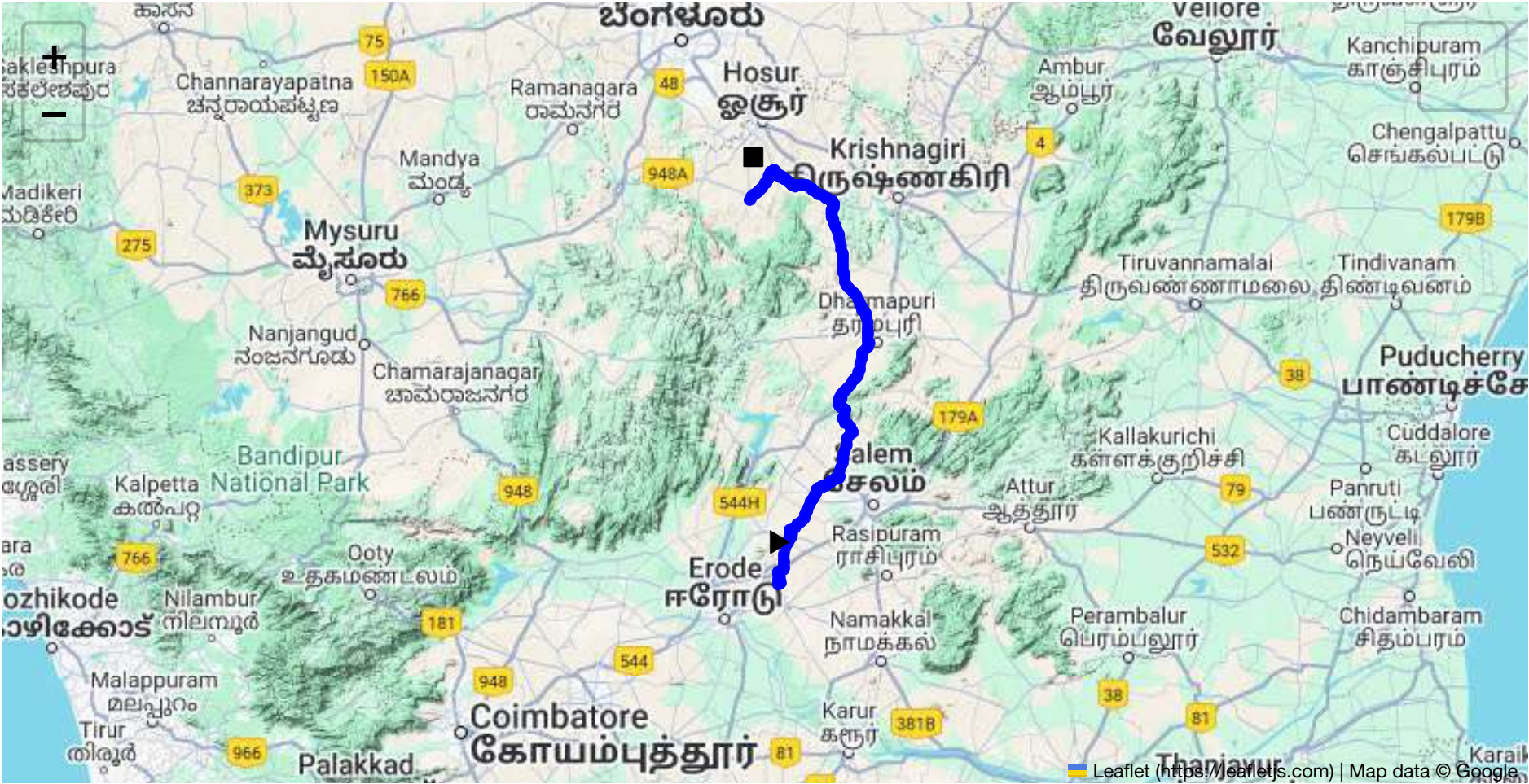
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 176.19 km
Estimated Duration: 3.7 hours
Adjusted Duration (Heavy Vehicle): 4.7 hours

Start: (11.4381, 77.8734)
End: (12.522869, 77.791557)



Welcome to the Journey Risk Management Study

Route Overview: The route from Sangagiri to Denkanikottai spans approximately 176.19 kilometers and passes through key locations such as Salem, Omalur, and Pillari Agraharam. It typically takes about 3.73 hours for heavy vehicles carrying hazardous materials, largely following national and state highways that connect various towns and districts in Tamil Nadu.

Weather Conditions: The climate in this region is typically tropical, with hot and dry conditions prevailing for much of the year. During the monsoon season (June to September), heavy rainfall can cause waterlogging and reduced visibility, leading to potential hazards. Additionally, during summer months, extreme heat can affect vehicle performance and driver fatigue.

Traffic Patterns and Congestion: Salem, being a key junction, experiences significant traffic, especially during peak hours from 8-10 AM and 5-8 PM. Highways connecting to Omalur and Pillari Agraharam are also known for congestion due to local traffic and intercity travel. Weekend travel can lead to more congestion, particularly near commercial zones and markets.

Road Quality and Infrastructure: The NH and SH routes primarily taken on this journey are generally well-maintained, though some segments near rural areas might have uneven surfaces or ongoing construction, potentially affecting travel time. Road signs are not consistently available in remote areas, requiring careful navigation.

Alternative Routes for Emergencies: During emergencies, bypassing Salem can be a viable alternative via rural roads, although these are narrower and less maintained. In severe weather conditions, opting for inland state roads that parallel major highways can provide alternatives but may increase travel time.

Local Regulations for Hazardous Material Transport: Tamil Nadu has strict regulations regarding the transport of hazardous materials, emphasizing proper labeling, containment, and the transporter's responsibility to notify authorities in case of any incidents. There are specific time restrictions in place to avoid peak traffic hours in urban areas.

Historical Incidents: There have been few documented incidents involving heavy vehicles in this region, primarily related to accidents at congested intersections or during adverse weather conditions. No significant hazardous material incidents have been noted, though vigilance in adherence to safety standards is advised.

Environmental Considerations: This route traverses near agricultural and residential areas, requiring caution to prevent spills or accidents. Proximity to water bodies near Omalur and Denkanikottai necessitates strict adherence to environmental guidelines to prevent contamination.

Communication Coverage: Mobile network coverage is generally reliable, but there may be potential dead zones in rural stretches between Salem and Denkanikottai. Ensuring multiple communication devices are available is advisable.

Emergency Response Times: Proximity to urban centers such as Salem allows for relatively quick emergency response times of approximately 30–45 minutes. However, in more remote stretches, this could extend up to 1–1.5 hours depending on terrain and accessibility.

Overall Risk Assessment Summary: The route presents moderate risks primarily due to weather conditions, traffic congestion in urban areas, and road quality variances. Adhering to safety protocols, using GPS navigation, and maintaining regular vehicle checks can mitigate many potential hazards. Close coordination with local authorities and emergency services is recommended to ensure swift action in case of incidents.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.60628, 77.86014	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	High	11.45028, 77.87202	15 KM/Hr
5	Turn	Medium	11.46400, 77.87323	30 KM/Hr
6	Turn	Medium	11.46765, 77.87037	30 KM/Hr
7	Turn	High	11.47403, 77.86841	15 KM/Hr
8	Turn	Medium	11.47986, 77.88397	30 KM/Hr
9	Turn	Medium	11.48363, 77.88780	30 KM/Hr
10	Turn	Medium	11.49108, 77.88784	30 KM/Hr
11	Turn	High	11.49451, 77.88489	15 KM/Hr
12	Turn	High	11.55593, 77.89757	15 KM/Hr
13	Turn	Medium	11.55644, 77.89922	30 KM/Hr
14	Turn	Medium	11.56644, 77.90654	30 KM/Hr
15	Turn	Medium	11.57693, 77.90864	30 KM/Hr
16	Turn	Medium	11.58172, 77.90856	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
17	Turn	Medium	11.58297, 77.90767	30 KM/Hr
18	Turn	High	11.58317, 77.90672	15 KM/Hr
19	Turn	Medium	11.59161, 77.90633	30 KM/Hr
20	Turn	Medium	11.59305, 77.90760	30 KM/Hr
21	Turn	Medium	11.63840, 77.94820	30 KM/Hr
22	Turn	High	11.64182, 77.95191	15 KM/Hr
23	Turn	Medium	11.64676, 77.95116	30 KM/Hr
24	Turn	Medium	11.67871, 77.96850	30 KM/Hr
25	Turn	Medium	11.69655, 77.97987	30 KM/Hr
26	Turn	High	11.70243, 77.97876	15 KM/Hr
27	Turn	Medium	11.71374, 77.99500	30 KM/Hr
28	Turn	Medium	11.71726, 77.99994	30 KM/Hr
29	Turn	Medium	11.71910, 78.01029	30 KM/Hr
30	Turn	Medium	11.73772, 78.03606	30 KM/Hr
31	Turn	High	11.73665, 78.03907	15 KM/Hr
32	Turn	Medium	11.74226, 78.04688	30 KM/Hr
33	Turn	Medium	11.78005, 78.05788	30 KM/Hr
34	Turn	Medium	11.86928, 78.08384	30 KM/Hr
35	Turn	Medium	11.91277, 78.05853	30 KM/Hr
36	Turn	Medium	11.93375, 78.06378	30 KM/Hr
37	Turn	Medium	11.93751, 78.05325	30 KM/Hr
38	Turn	Medium	11.95551, 78.04706	30 KM/Hr
39	Turn	High	11.95865, 78.04991	15 KM/Hr
40	Turn	High	11.96021, 78.04945	15 KM/Hr
41	Turn	Medium	11.96281, 78.05124	30 KM/Hr
42	Turn	High	11.96442, 78.05104	15 KM/Hr
43	Turn	High	11.96557, 78.04930	15 KM/Hr
44	Turn	Medium	12.12041, 78.13284	30 KM/Hr
45	Blind Spot	Blind Spot	12.51381, 78.03208	10 KM/Hr
46	Turn	Medium	12.51395, 78.02897	30 KM/Hr
47	Turn	High	12.51632, 78.02571	15 KM/Hr
48	Turn	Medium	12.52154, 78.02425	30 KM/Hr
49	Turn	Medium	12.52683, 78.00972	30 KM/Hr
50	Turn	High	12.55119, 77.97237	15 KM/Hr
51	Turn	Medium	12.55300, 77.96561	30 KM/Hr
52	Turn	Medium	12.56463, 77.95429	30 KM/Hr
53	Turn	Medium	12.56244, 77.92201	30 KM/Hr
54	Turn	Medium	12.57383, 77.89811	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
55	Turn	Medium	12.58265, 77.89337	30 KM/Hr
56	Turn	Medium	12.58407, 77.88828	30 KM/Hr
57	Turn	Medium	12.58361, 77.88737	30 KM/Hr
58	Turn	Medium	12.58526, 77.88154	30 KM/Hr
59	Turn	Medium	12.59185, 77.86994	30 KM/Hr
60	Turn	Medium	12.59956, 77.86307	30 KM/Hr
61	Turn	High	12.60468, 77.85443	15 KM/Hr
62	Turn	Medium	12.60203, 77.85299	30 KM/Hr
63	Turn	Medium	12.59337, 77.84801	30 KM/Hr
64	Turn	Medium	12.55183, 77.81689	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
6	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
7	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
8	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
9	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
10	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
11	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
12	hospital	Saravana Hospital	11.7409303,	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
			78.0458308		
13	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
14	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
15	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
16	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
17	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
19	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
20	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
21	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
22	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
23	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
24	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
25	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
26	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
27	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
29	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
30	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
31	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
33	hospital	Government Hospital, Jakkasamudram	12.3937246, 78.0522641	30 km/h	Medium
35	clinic	Kowthalam HSC	12.5676496, 77.8312573	30 km/h	Medium
37	police	Denkanikottai Police Station	12.5208971, 77.7903372	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
18	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
28	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
32	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
34	marketplace	Rayakottai tomato market	12.5237469, 78.0220377	30 km/h	Medium
36	school	St. Jopseph School	12.5339473, 77.795134	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.60628, 77.86014



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



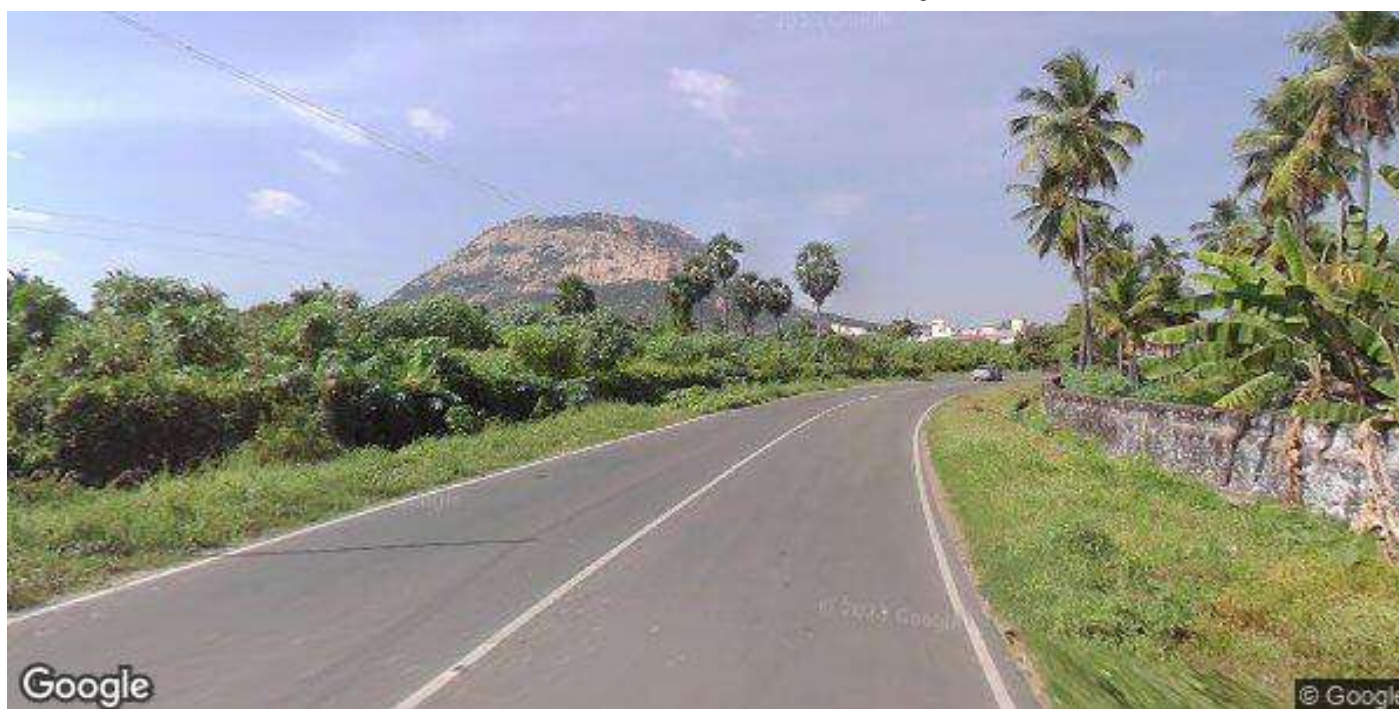
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47986, 77.88397



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48363, 77.88780



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49108, 77.88784



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49451, 77.88489



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55593, 77.89757



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58317, 77.90672



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59161, 77.90633



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59305, 77.90760



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63840, 77.94820



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64182, 77.95191



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64676, 77.95116



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67871, 77.96850



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69655, 77.97987



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70243, 77.97876



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71374, 77.99500



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71726, 77.99994



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71910, 78.01029



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73772, 78.03606



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73665, 78.03907



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74226, 78.04688



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78005, 78.05788



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86928, 78.08384



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91277, 78.05853



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93375, 78.06378



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93751, 78.05325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95551, 78.04706



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95865, 78.04991



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96021, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96281, 78.05124



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96442, 78.05104



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96557, 78.04930



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12041, 78.13284



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.51381, 78.03208



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51395, 78.02897



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.51632, 78.02571



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52154, 78.02425



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52683, 78.00972



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.55119, 77.97237



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.55300, 77.96561



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.56463, 77.95429



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.56244, 77.92201



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.57383, 77.89811



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58265, 77.89337



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58407, 77.88828



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58361, 77.88737



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58526, 77.88154



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59185, 77.86994



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59956, 77.86307



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.60468, 77.85443



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.60203, 77.85299



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59337, 77.84801



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.55183, 77.81689
